

Statement of Purpose

Applicant: Fuad Ahmed | **Nationality:** Ethiopian | **Degree:** BSc in Computer Science
Target Program: Open Doors: Russian Scholarship Project (Association of Global Universities)
Subject Track: Computer and Data Science (Master's Level) | **Preferred Universities:** ITMO, Innopolis, HSE

1. INTRODUCTION & ACADEMIC OBJECTIVE

I am writing to express my focused motivation to pursue a Master of Science in Computer and Data Science through the Open Doors Russian Scholarship Project. Having graduated with a Bachelor of Science in Computer Science from St. Mary's University in Addis Ababa with a 3.20 Major GPA, I have spent the last several years architecting, building, and deploying real-world software systems. My academic objective is to advance my technical capabilities in distributed computing, concurrency control mechanisms, and scalable system architectures under the mentorship of Russia's leading technical faculties.

2. ACADEMIC BACKGROUND & FOUNDATIONS

Throughout my undergraduate education, I gravitated toward the structural and engineering aspects of computing. I achieved top marks across core software engineering disciplines, including Advanced Object-Oriented Programming (A+), Web & Internet Programming (A/A), Relational Database Design (A), and Senior Software Capstone Projects (A/A+). In addition to my primary computer science curriculum, I completed intensive professional training in Capital Markets & Securities Analysis at the Addis Ababa University School of Commerce, which deepened my understanding of transaction integrity, market structures, and the rigorous mathematical standards demanded by financial platforms.

3. ENGINEERING PRACTICE & SYSTEMS PROBLEM SOLVING

Rather than treating software engineering as theoretical exercises, my focus has consistently been on engineering software that operates reliably in constrained, high-stakes environments:

- Transactional Integrity & Concurrency Architecture:** In developing *LuckyDraw*, a high-volume Telegram Mini App lottery engine built on Laravel and Livewire, I addressed race conditions and double-spending risks by enforcing atomic row-locking mechanisms (`DB::transaction` with `lockForUpdate`) and strict integer-cent financial ledger arithmetic (`App\Support\Money`). To validate system resilience under high concurrency, I engineered automated test suites covering deadlock avoidance, stuck-draw recovery, and HMAC-authenticated payload verification.
- Geospatial Systems & Crowdsourced Consensus:** In creating *Addis Condo Finder*, I built a cross-platform Flutter application and Laravel backend to map complex condominium developments across Addis Ababa. I implemented bounding-box viewport queries, offline-first local caching, and a multi-user verification pipeline requiring independent consensus before crowdsourced

geospatial coordinates are published.

3. **Android Systems & Low-Level Lifecycle:** Through direct interaction with mobile operating system internals, I have engineered persistent Android Foreground Services, inter-process communication (IPC) routines, and battery-optimization bypass mechanisms, ensuring high-frequency background synchronization across real-world field devices.

4. WHY RUSSIA & TARGET UNIVERSITIES

Russian technical institutions are globally recognized as the apex of algorithmic rigor, competitive programming, and systems engineering. I am specifically targeting:

- **ITMO University (St. Petersburg):** As a seven-time champion of the ICPC World Finals, ITMO represents the highest international standard in algorithm design, distributed computing, and system optimization.
- **Innopolis University (Kazan):** As Russia's premier specialized IT campus, Innopolis provides an English-medium environment deeply intertwined with industry-grade software architecture, AI, and robotics.
- **HSE University (Faculty of Computer Science, Moscow):** Renowned for its advanced research partnerships with industry leaders like Yandex in large-scale data engineering and distributed systems.

5. LONG-TERM VISION & IMPACT

The Horn of Africa is undergoing a rapid digital transformation, driven by mobile payment platforms (such as Telebirr, CBE Birr, and Chapa) and massive urban logistics systems. However, a major bottleneck remains the scarcity of engineers trained in deep distributed systems, fault-tolerant backend architectures, and high-concurrency data pipelines. By completing a Master of Science in Russia, I intend to bridge the gap between academic systems research and applied engineering, building robust, fault-tolerant digital infrastructure that supports millions of users. The Open Doors Scholarship represents the definitive catalyst for this mission, and I am prepared to contribute my work ethic, systems experience, and dedication to your academic community.